Name parts of circles

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________________ Date: _______________

1. Read the model: A circle has radius 4.5 cm. What is its diameter? Copy or complete the first step.

2. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

3. Angles around a point are 85 degrees, 140 degrees and x. Find x.

4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A circle has radius 4.5 cm. What is its diameter? Copy or complete the first step.
   Answer 9 cm.

2. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
   Answer 75 degrees.

3. Angles around a point are 85 degrees, 140 degrees and x. Find x.
   Answer 135 degrees.

4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.
   Answer Use labelled values and known geometric facts, then check the total angle relationship.

Name parts of circles

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A circle has radius 4.5 cm. What is its diameter?

2. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

3. Angles around a point are 85 degrees, 140 degrees and x. Find x.

4. Solve this independently and show every stage: A circle has radius 4.5 cm. What is its diameter?

5. Use a second method or representation for: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

6. Check this result using an inverse, estimate or known fact: Angles around a point are 85 degrees, 140 degrees and x. Find x.

7. Prove or explain your reasoning: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Angles around a point are 85 degrees, 140 degrees and x. Find x.

Teacher answers and checking prompts

1. A circle has radius 4.5 cm. What is its diameter?
   Answer 9 cm.

2. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
   Answer 75 degrees.

3. Angles around a point are 85 degrees, 140 degrees and x. Find x.
   Answer 135 degrees.

4. Solve this independently and show every stage: A circle has radius 4.5 cm. What is its diameter?
   Answer 9 cm.

5. Use a second method or representation for: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
   Answer Expected result: 75 degrees. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Angles around a point are 85 degrees, 140 degrees and x. Find x.
   Answer Expected result: 135 degrees. A valid check must be shown.

7. Prove or explain your reasoning: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
   Answer 68 degrees. The explanation must show why the method works.

8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.
   Answer Use labelled values and known geometric facts, then check the total angle relationship.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Angles around a point are 85 degrees, 140 degrees and x. Find x.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Name parts of circles

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ______________

1. Prove or explain your reasoning: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

2. Prove the answer to this question, rather than only calculating it: A circle has radius 4.5 cm. What is its diameter?

3. Solve this in two different ways and compare them: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

4. Work backwards from the answer to reconstruct the question: 135 degrees.

5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.

6. Create a new example that tests the same idea as: Angles around a point are 85 degrees, 140 degrees and x. Find x.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
   Answer 68 degrees. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A circle has radius 4.5 cm. What is its diameter?
   Answer Expected result: 9 cm. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
   Answer Expected result: 75 degrees. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 135 degrees.
   Answer One valid reconstruction is: Angles around a point are 85 degrees, 140 degrees and x. Find x. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.
   Answer Use labelled values and known geometric facts, then check the total angle relationship.

6. Create a new example that tests the same idea as: Angles around a point are 85 degrees, 140 degrees and x. Find x.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.